OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
TECHNOLOGICAL UNIVERSITY DUBLIN

TU850/2

TEAM PROJECT: DATA PROCESSING AND MANIPULATION

DATA 2005 - Data-Centric Programming
CA1
Presentation

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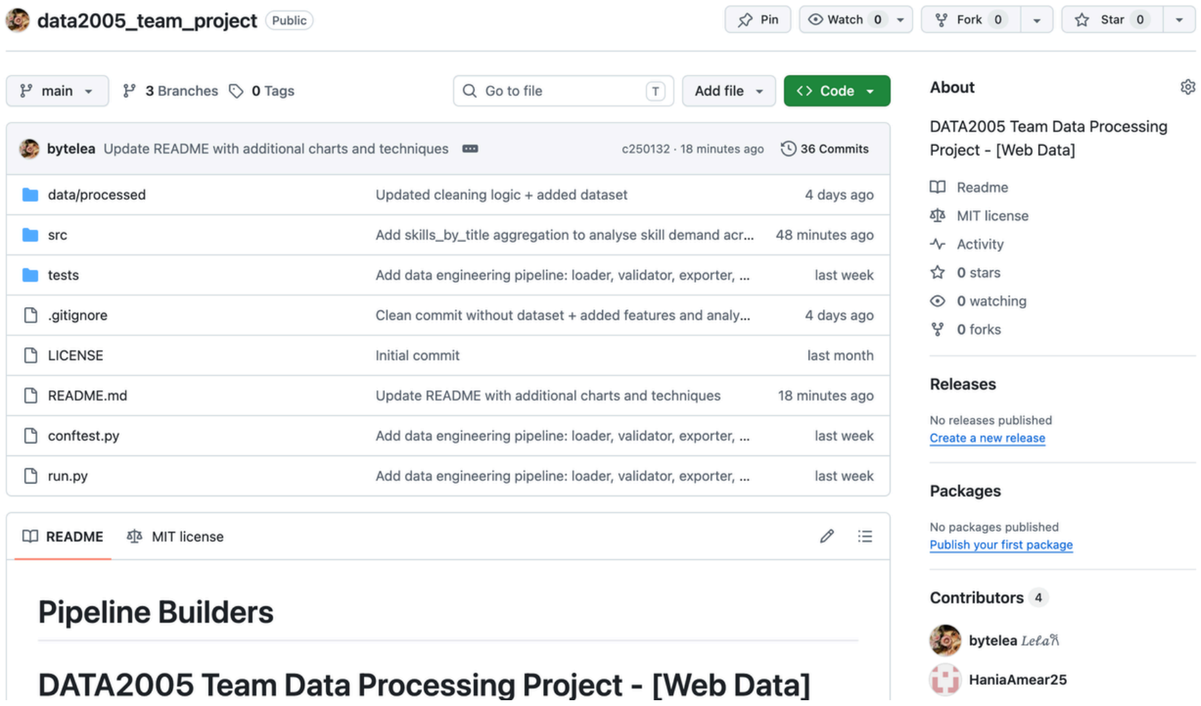
Introduction

- Web-based job market analysis focused on Data Analyst roles in the US
- Goal: uncover trends in skills, salaries, locations and job titles
- Built a full data processing pipeline in Python

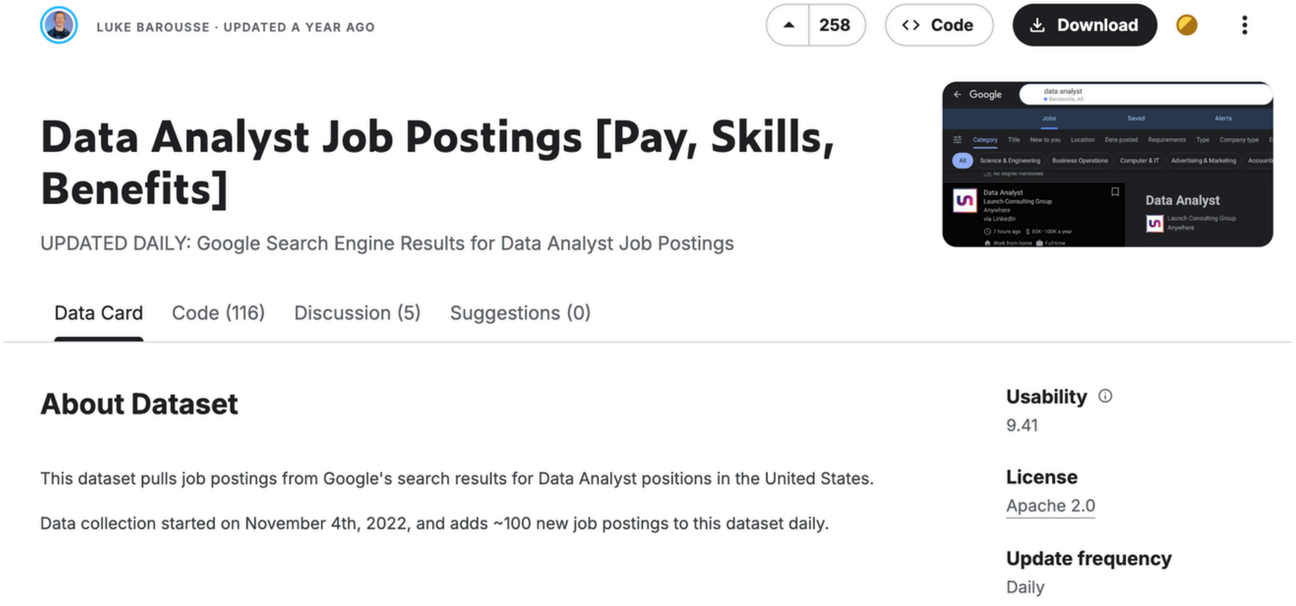
Github



- Repository: github.com/bytelea/data2005_team_project
- Structure: src/pipeline/ contains all modules
- Entry point: python run.py runs the full pipeline

Dataset

- Name: Data Analyst Job Postings [Pay, Skills, Benefits]
- Source: Kaggle - Luke Barousse
- Collected from: Google Search job results
- Size: ~10,000+ records
- Format: CSV
- Key variables: job title, company, location, salary, skills, remote status, seniority

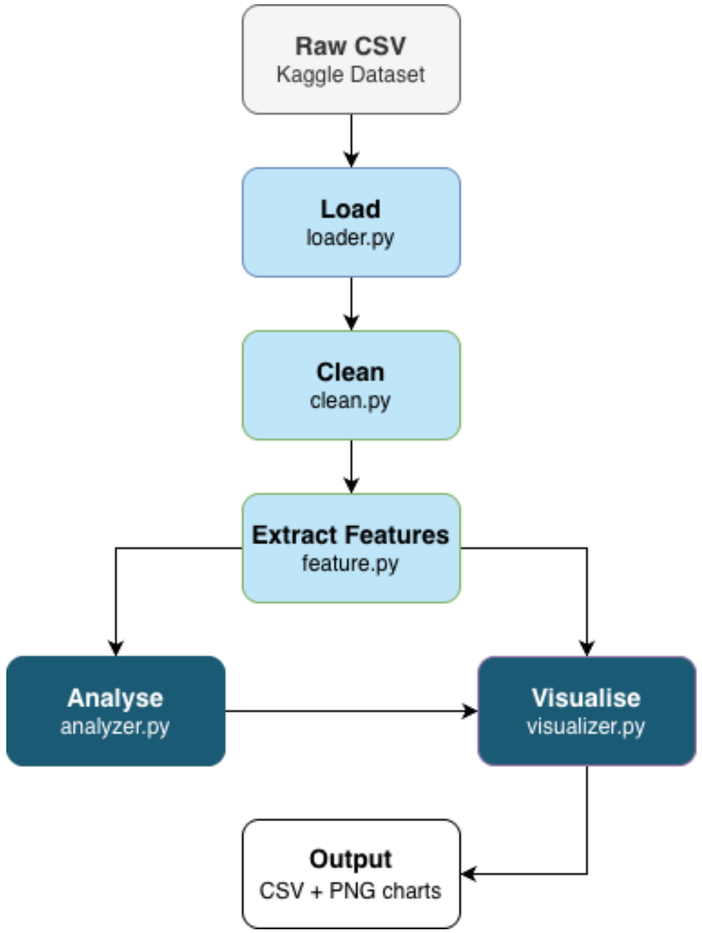


How it Works

```python
# STEP 5 - Analyse trends  (Data Analyst)
# Calculates top skills, salaries, job counts using analyzer.py
print("\n[Step 5/6] Analysing trends...")
results = analyze(df, top_n=self.top_n)


# STEP 6 - Visualise and export  (Visualization Lead + Data Engineer)
# Saves charts using visualizer.py, saves CSV files using exporter.py
print("\n[Step 6/6] Saving charts and files...")
if self.charts:
    try:
        visualize(results)
    except ImportError:
        print("[Visualizer] matplotlib/seaborn not installed — skipping charts.")
export(df, results)

print("\n" + "=" * 60)
print(" PIPELINE COMPLETE")
print("=" * 60 + "\n")

return df, results
```

```python
# STEP 1 - Load data  (Data Engineer)
# Reads the CSV or JSON file into a table using loader.py
print("\n[Step 1/6] Loading data...")
df = load_data(filepath)
print(f"          {len(df):,} rows x {len(df.columns)} columns loaded.")


# STEP 2 - Validate data  (Data Engineer)
# Checks for missing columns, empty values, duplicates using validator.py
print("\n[Step 2/6] Validating...")
report = validate_data(df)
if self.verbose:
    print(report.summary())
if not report.passed and self.strict:
    raise PipelineError("Validation failed. Fix errors before continuing.\n" + report.summary())


# STEP 3 - Clean data  (Data Analyst)
# Removes duplicates, fills missing values, parses skills using cleaner.py
print("\n[Step 3/6] Cleaning...")
df = clean_data(df)
print(f"          {len(df):,} rows after cleaning.")
```

Data Processing Pipeline



Data Cleaning

- Removed duplicate rows
- Stripped whitespace from text columns
- Filled missing salary values with median
- Parsed skill strings into Python lists

```python

    # Check that all required columns exist
    for col in REQUIRED_COLUMNS:
        if col.lower() not in lower_cols:
            report.errors.append(f"Missing required column: '{col}'")

    # Check that the dataset has at least one row
    if len(df) == 0:
        report.errors.append("The dataset has 0 rows.")

    # Stop here if there are already errors
    if not report.passed:
        return report

    # Check that critical columns have no empty values
    for col in CRITICAL_COLUMNS:
        actual = lower_cols.get(col.lower())
        if actual:
            null_count = df[actual].isna().sum()
            if null_count > 0:
```

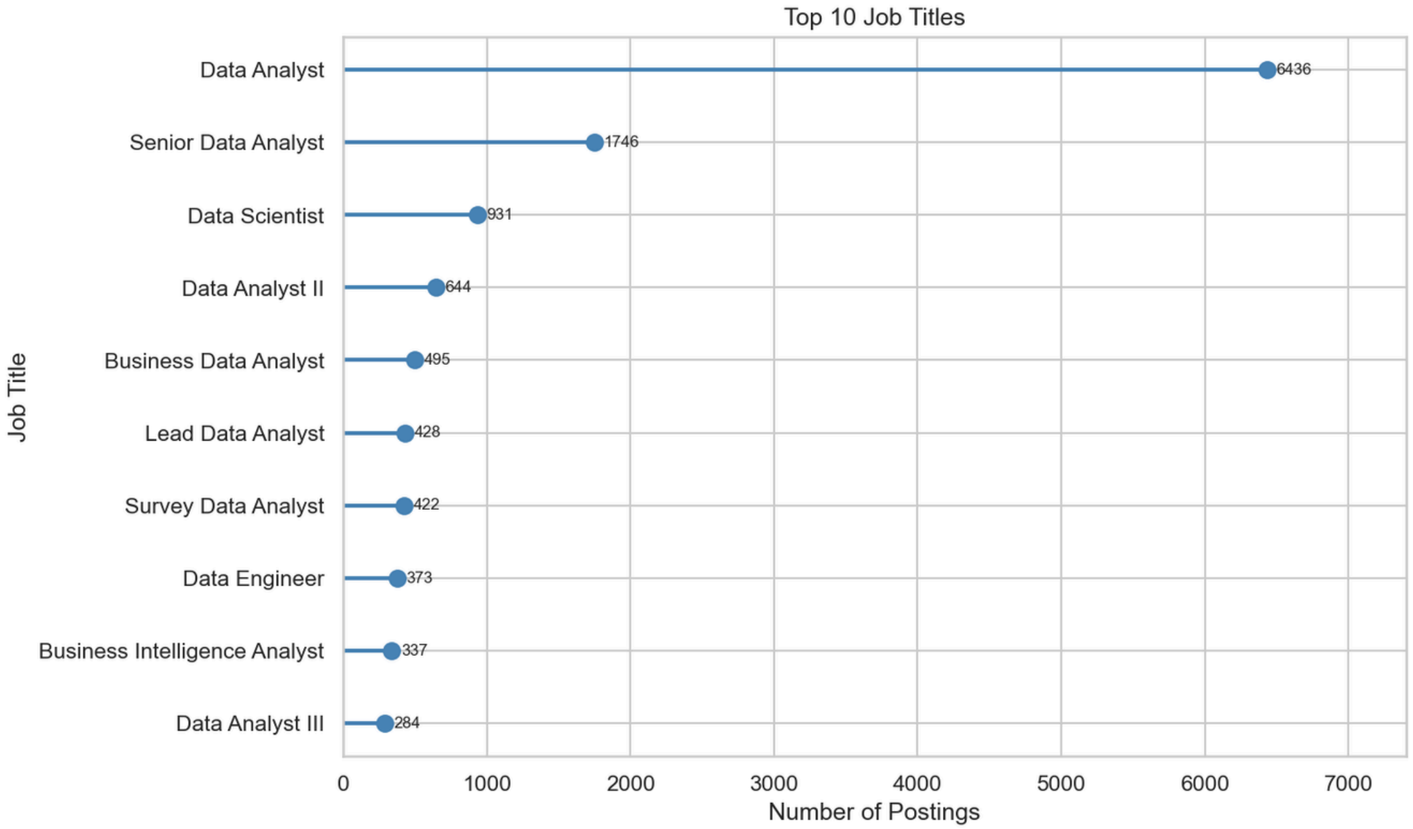
Data Analysis

- Identified top hiring companies using frequency analysis
- Calculated remote work percentage using work_from_home column
- Extracted in-demand skills using str.contains() on job descriptions
- Created boolean columns for Python, SQL, Excel, Tableau, Power BI
- Computed summary statistics using NumPy (.mean(), .sum())
- Grouped data by job title and state to find patterns

Key Findings & Visualisations

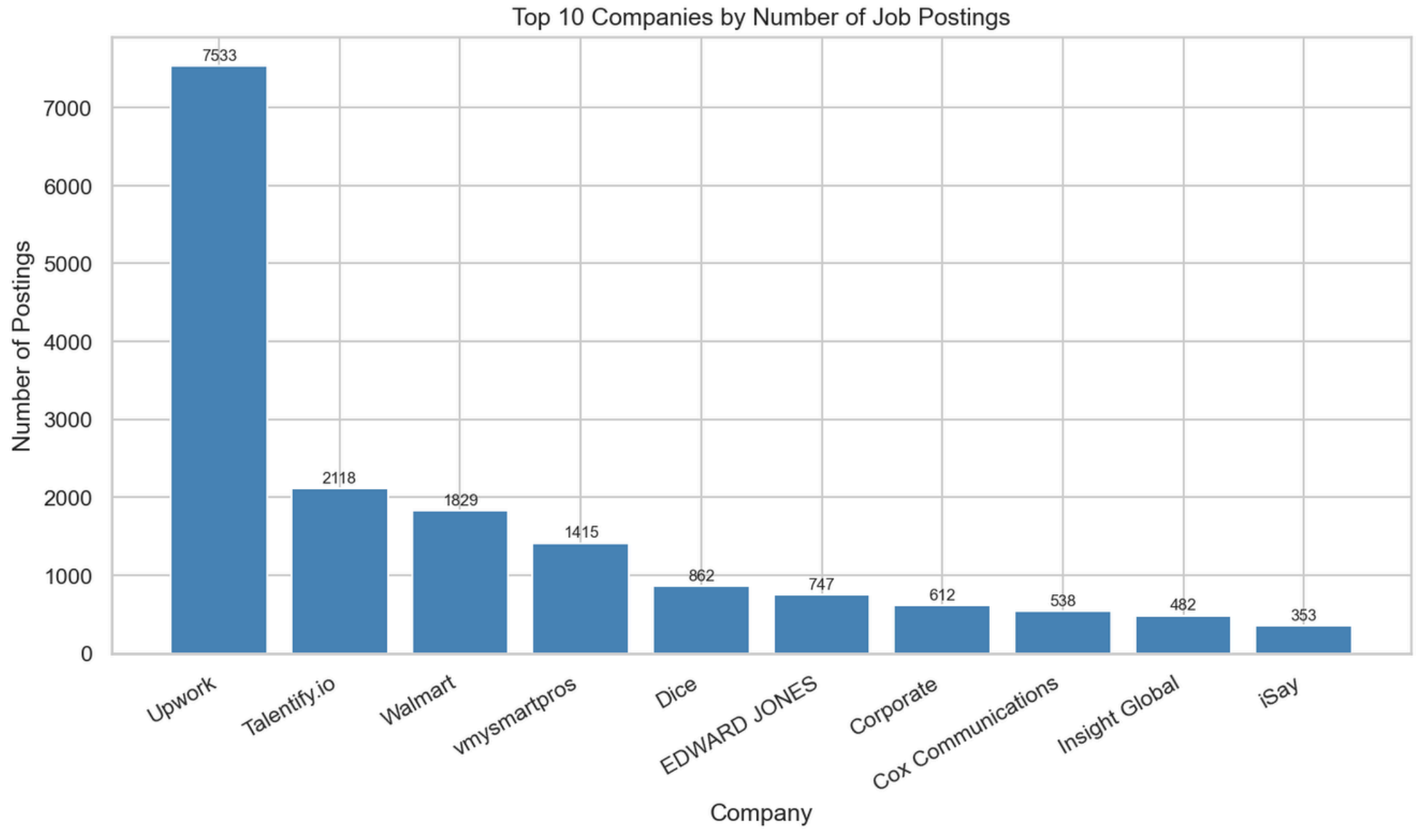
- Excel & SQL are the most in-demand skills overall
- Data Analyst role dominates the market (6,436 postings)
- The job market is overwhelmingly remote-first
- Upwork accounts for a massive share of all postings
- Python is critical for Data Scientists specifically

Key Findings & Visualisations



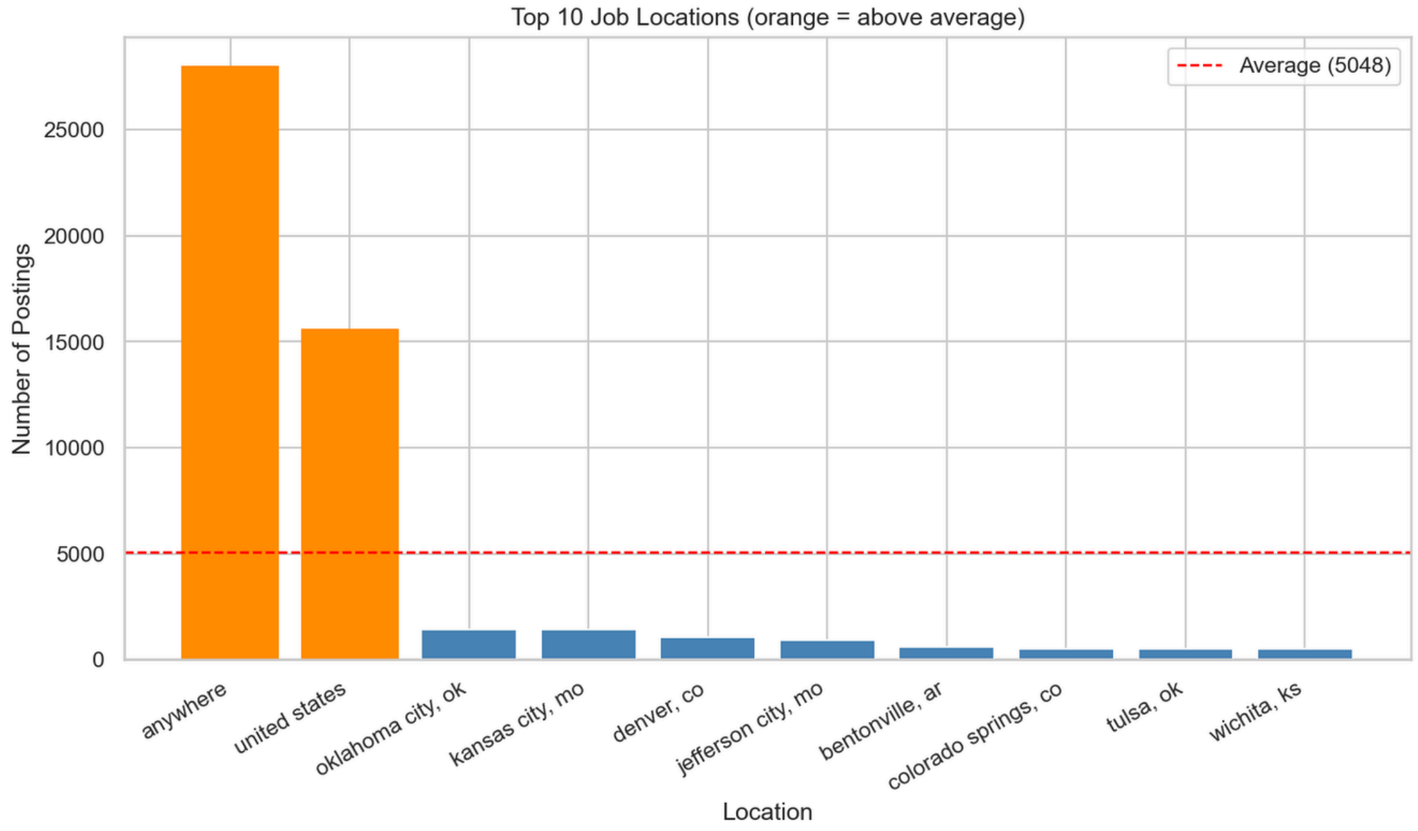
Data Analyst is #1 role - 6,436 postings

Key Findings & Visualisations



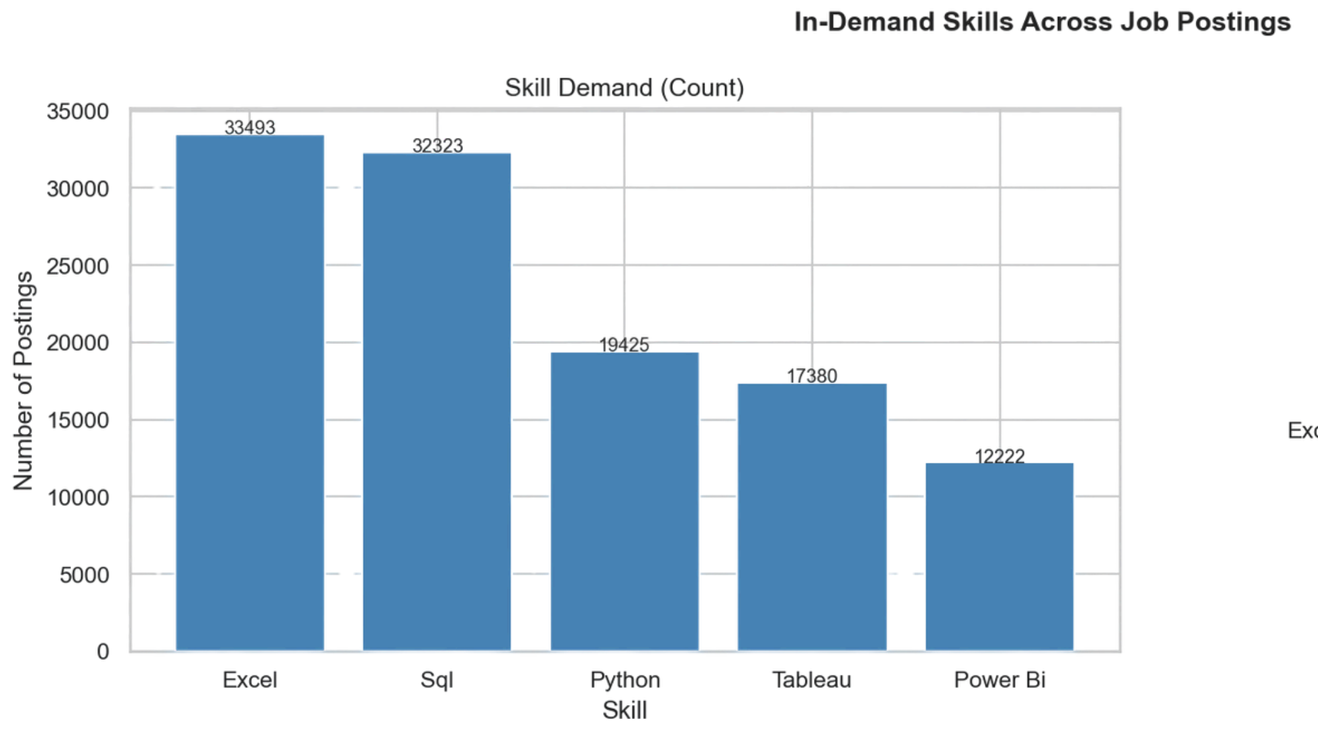
Upwork leads hiring with 7,533 job posts

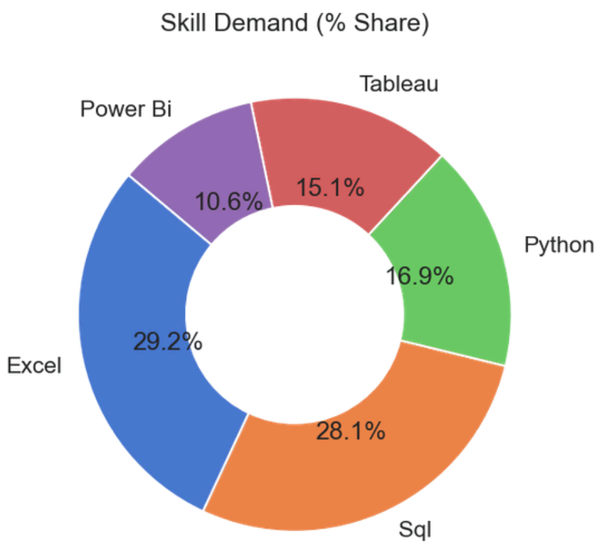
Key Findings & Visualisations



40,000+ roles listed as "Anywhere" - remote first market

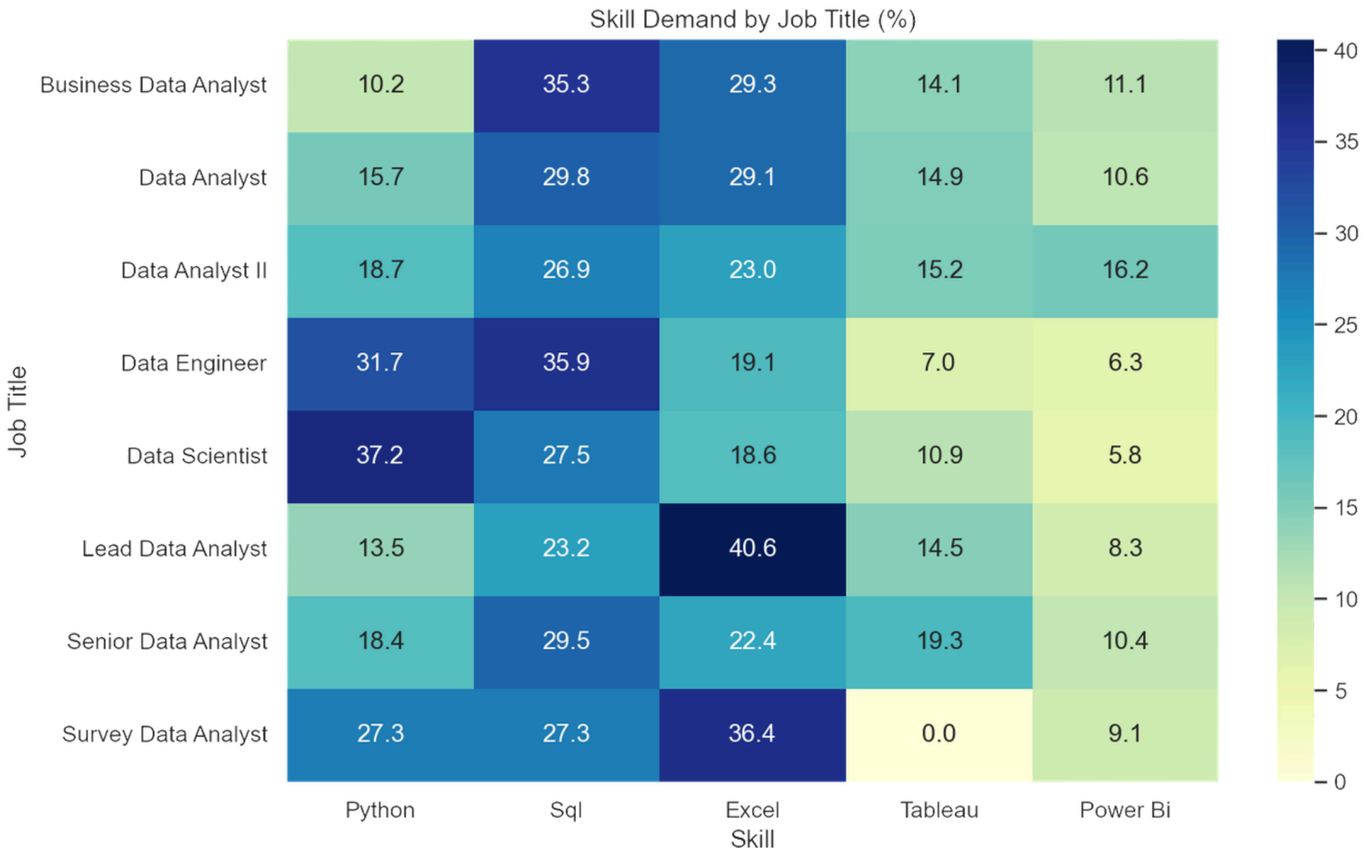
Key Findings & Visualisations





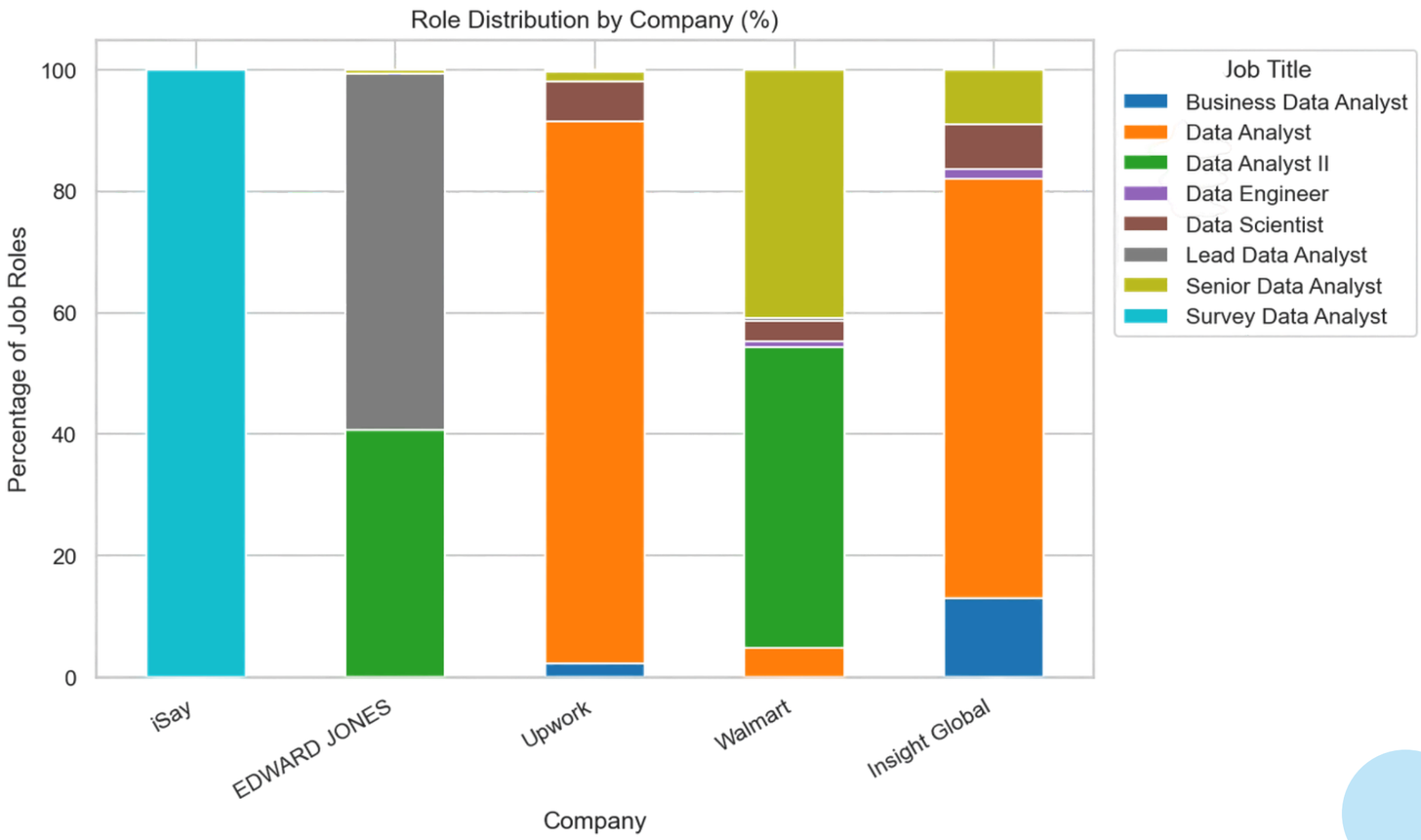
Excel most demanded for Lead roles (40.6%)

Key Findings & Visualisations



Python highest for Data Scientists (37.2%)

Key Findings & Visualisations



SQL consistently demanded across all roles

Challenges & Lessons Learned

- Parsing raw skill strings into usable lists was complex
- Missing salary data required median imputation decisions
- Managing Git across 4 people taught us real workflow skills
- Dataset was web-sourced but not a traditional domain dataset
- Balancing different coding styles across modules

Thank you for listening